

Legend:
Light Blue extractStructuredDom(url)
Dark Yellow waitForCmpUi(page, selectorMap, timeout = 10000)
Light Green findCorrectFrame(page, selectorMap)
Light Violet frameWordCounter(frames)
Light Red calculateFrameScore(frame, avgWordCount, selectorMap, iFrameBonus)
Light Yellow extractFromFrame(frame, selectors, selectorsMap, cmpType = null)
Red clickAndExtractSettings(frame, selector, page, cmpType)
Blue getFrameState(frame)
Gray findSettingsButtonViaLlm(html)

Additional Hints for Used Variables

- page:** The active Puppeteer page instance representing the current browser tab and session.
- frame:** Any browsing context on the page as represented by Puppeteer's Frame API, either the main document or one of its embedded frames.
- CMP_SELECTORS_MAP:** A comprehensive dictionary mapping known CSS selectors to their respective Consent Management Platform (CMP) provider names (e.g., "OneTrust").
- CMP_SELECTORS:** An array containing only the CSS selector keys from CMP_SELECTORS_MAP.
- GENERAL_SELECTORS:** An array of general CSS selectors for detecting cookie consent dialog candidates.
- cmpType:** The specific name of the identified CMP (e.g., "Sourcepoint").
- selector:** A precise CSS string (such as an ID or an aria-label) used to target a specific interactive element, e.g. a settings button or cookie banner container.
- isMainFrame:** Boolean flag indicating whether this frame is the page's main document (frame === page.mainFrame()), as opposed to an embedded iframe.
- isCookieBannerFrame:** Boolean flag indicating whether this frame is considered the identified consent banner frame. Set to true unconditionally when a frame was selected via the scoring heuristic (findCorrectFrame); when no frame scored above threshold and all frames are processed as a fallback, this flag instead reflects whether the extraction itself detected a known CMP container within that specific frame.
- SETTINGS_TERMS_REGEX:** A regex instance of an array including 1280 setting-button phrases.

- Frame URL matching with CMP_DOMAINS array → +50
- Frame URL/name CMP regex matching (CMP_FRAME_REGEX) → +20
- visibility check (display:none/hidden, zero dimensions) → return -100 (scoring stops)
- word count < 20 → -20
- word count > avg + 100 → -20
- word count = 0 → -100
- General CSS selector match (GENERAL_SELECTORS) → +5
- CMP-specific selector match (CMP_SELECTORS_MAP) → +10
- N-gram match (unigram +1 ... 5-gram +5)
- trigger word match (multilingual consent vocabulary) (+2 for each, capped at +10)
- fixedHigh-z element inside frame (excl. revoke buttons <100px and nav elements with >10 internal links) → +15;
- if also displayed at top of screen → additional +5
- return score

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM

- For each button/checkbox/toggle element, information like text, attributes and selector/selectorConfidence gets stored (the full overview is visible in the code)
- Utilizing helper functions, which are also functional for the shadow DOM